



Outline

Probabilities

Entropy and Information

Sampling Methods

- Random Sampling
- Stratified Sampling
- Systematic Sampling
- Cluster Sampling
- Bootstrap Sampling
- Oversampling & Undersampling



Outline

Probabilities

Entropy and Information

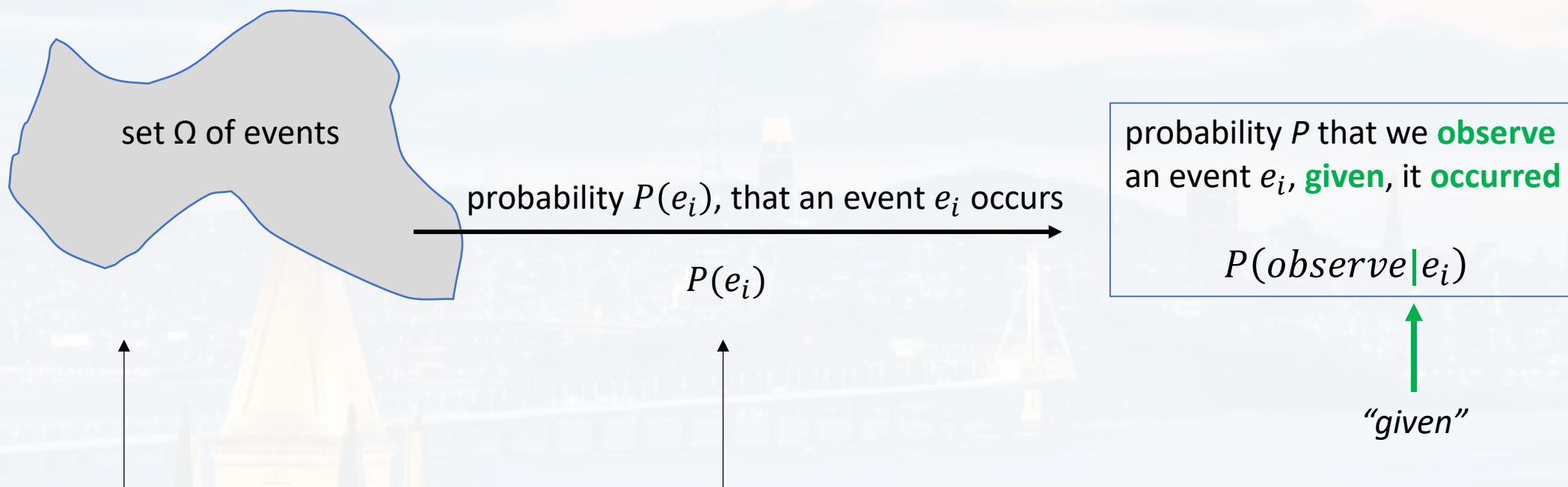
Sampling Methods

- Random Sampling
- Stratified Sampling
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- Cluster Sampling
- Bootstrap Sampling
- Oversampling & Undersampling



here: **heuristic explanation** → more mathematical rigorous: see **Cox's theorem**

axioms of probability
important quantities
the sampling error



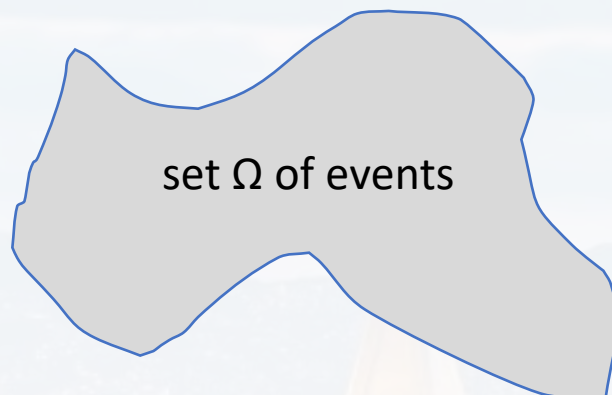
from data/observations → deriving Ω

from data/observations → understanding process
→ finding $P(e_i)$



here: **heuristic explanation** → more mathematical rigorous: see **Cox's theorem**

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from data/observations

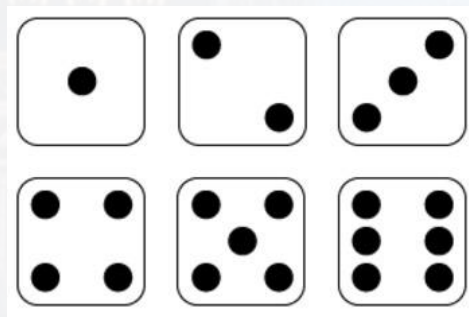
→ deriving Ω

1 2 2 1 2 2 1 2 1 1 2 1 1 2

observations

Is the observation 3 just rare (**and that's why we haven't observed it**), or is $3 \notin \Omega$?

events can be **discrete**



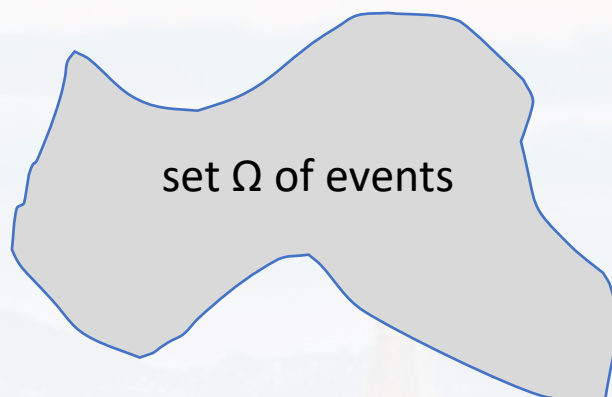
or **continuous**

- car speed measured in a speed trap
- a person's weight, etc



$P(e_i)$, that an event e_i occurs

axioms of probability
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1st axiom:

$P(e_i)$ is a **non-negative, real number**

2nd axiom:

the probability that at **least one** of the events in the entire sample space will occur is **1**
if events are **collectively exhaustive**

from 1st and 2nd : $P(e_i) = [0, 1]$ for any e_i

If events are **mutually exclusive**: 3rd axiom:

$$P(\bigcup_{i=1}^{\infty} e_i) = \sum_{i=1}^{\infty} P(e_i)$$

$\cup$: “union”

$\bigcup_{i=1}^{\infty} e_i$ means e_1 **or** e_2 **or** e_{∞}



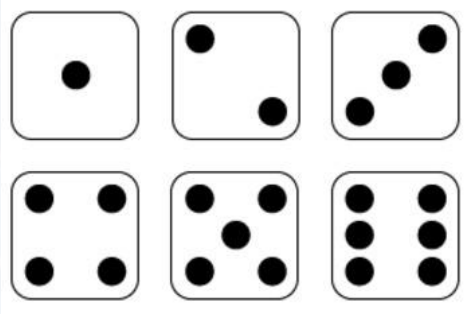
$P(e_i)$, that an event e_i occurs

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If events are **mutually exclusive**: 3rd axiom: $P(\bigcup_{i=1}^{\infty} e_i) = \sum_{i=1}^{\infty} P(e_i)$

U: “union”

$\bigcup_{i=1}^{\infty} e_i$ means e_1 **or** e_2 **or** e_{∞}



The probability that we roll a **4 or a 6** equals...

$$P\left(e_4 \bigcup e_6\right) =$$

...the **probability** that we roll a **4 plus** the **probability** that we roll a **6**

$$P(e_4) + P(e_6)$$

“or” equals addition!



$P(e_i)$, that an event e_i occurs

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“or” equals addition!

$$P(\cup_{i=1}^{\infty} e_i) = \sum_{i=1}^{\infty} P(e_i)$$

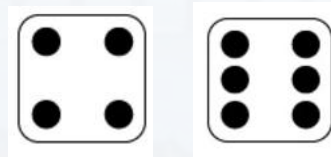
$\cup$: “union”

“and” equals multiplication!

$$P\left(\bigcap_{i=1}^{\infty} e_i\right) = \prod_{i=1}^{\infty} P(e_i)$$

two dice: The probability that we roll a **4 and a 6** equals...

$$P\left(e_4 \cap e_6\right) =$$



$\cap$: “intersection”

...the **probability** that we roll a **4 times** the **probability** that we roll a **6**

$$P(e_4)P(e_6)$$



“or” equals addition!

$$P(\cup_{i=1}^{\infty} e_i) = \sum_{i=1}^{\infty} P(e_i)$$

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“and” equals multiplication!

$$P\left(\bigcap_{i=1}^{\infty} e_i\right) = \prod_{i=1}^{\infty} P(e_i)$$

$\cup$: “union”

$\cap$: “intersection”

Be careful if events are not mutually exclusive (like a **set of events** or a **sequence of events**)

two light bulbs A and B:

What is the probability
that A or B is turned on?



A



B



“or” equals addition!

$$P(\cup_{i=1}^{\infty} e_i) = \sum_{i=1}^{\infty} P(e_i)$$

axioms of probability
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“and” equals multiplication!

$$P\left(\bigcap_{i=1}^{\infty} e_i\right) = \prod_{i=1}^{\infty} P(e_i)$$

$\cup$: “union”

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Be careful if events are not mutually exclusive (like a **set of events** or a **sequence of events**)

two light bulbs A and B:



A



B

What is the probability
that A **or** B is turned on?

It depends on how you define
“or”



two light bulbs A and B: **EX**clusive **OR**

axioms of probability
important quantities
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A



B



A



B



$$P(A) + P(B)$$

U: "union"

∩: "intersection"



A



B



A



B



$$P(A)P(B)$$

$$P\left(A \cup B\right) = P(A) + P(B) - 2P\left(A \cap B\right)$$

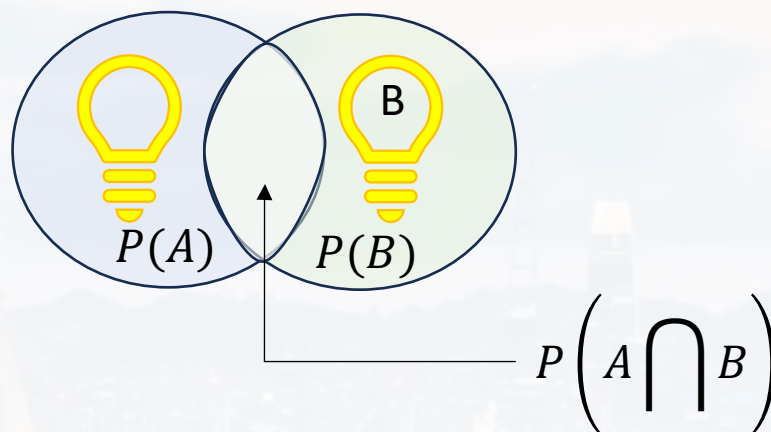
$$= P(A) + P(B) - 2P(A)P(B)$$



two light bulbs A and B: **EX**clusive **OR**

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Venn diagram:



Note: Be careful with Venn diagrams!
They are not a mathematical proof!

U: “union”

$\cap$: “intersection”

size of **blue** area: $P(A)$

size of **green** area: $P(B)$

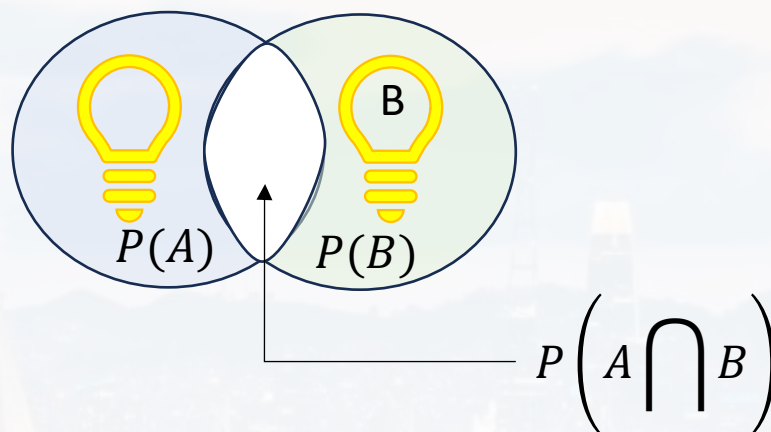
size of **blue** + **green** area: $P(A) + P(B) - P(A \cap B)$



two light bulbs A and B: **EX**clusive **OR**

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Venn diagram:



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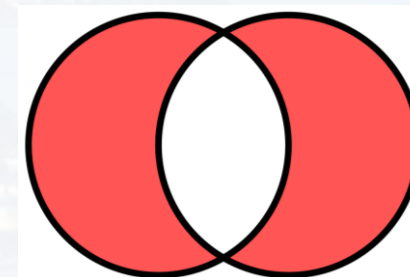
size of **blue** area: $P(A)$

size of **green** area: $P(B)$

size of **blue** + **green** area: $P(A) + P(B) - P(A \cap B)$

EXclusive **OR**: removing the overlapping part → making a “hole”

$$P(A \cup B) = P(A) + P(B) - 2P(A \cap B)$$

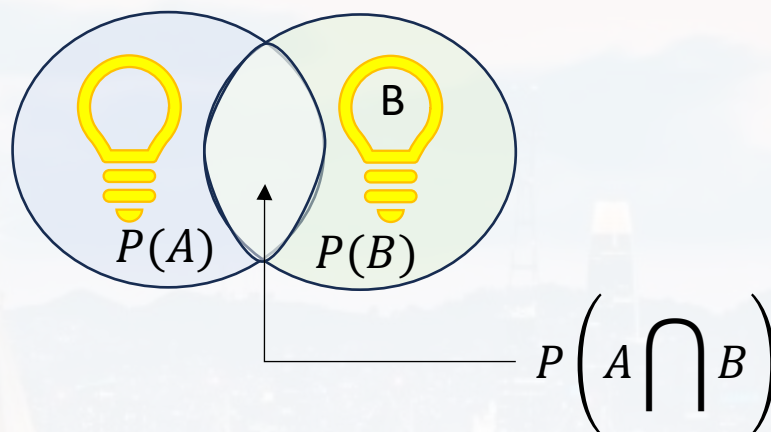




two light bulbs A and B: **IN**clusive **OR**

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Venn diagram:



Note: Be careful with Venn diagrams!
They are not a mathematical proof!

U: “union”

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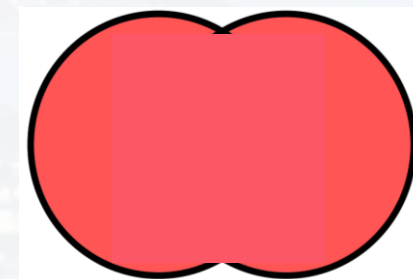
size of **blue** area: $P(A)$

size of **green** area: $P(B)$

size of **blue** + **green** area: $P(A) + P(B) - P(A \cap B)$

Inclusive OR: no “hole”

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$





two light bulbs A and B: **IN**clusive **OR**

axioms of probability
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A



B



A



B



$$P(A) + P(B)$$

U: “union”

∩: “intersection”



A



B



A



B



$$P(A)P(B)$$

$$P\left(A \cup B\right) = P(A) + P(B) - P\left(A \cap B\right)$$

$$= P(A) + P(B) - P(A)P(B)$$



“or” equals addition!

$$P(\cup_{i=1}^{\infty} e_i) = \sum_{i=1}^{\infty} P(e_i)$$

axioms of probability
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“and” equals multiplication!

$$P\left(\bigcap_{i=1}^{\infty} e_i\right) = \prod_{i=1}^{\infty} P(e_i)$$

$\cup$: “union”

$\cap$: “intersection”

inclusion - exclusion principle (here: INclusive OR)

$$P\left(e_4 \cup e_6\right) = P(e_4) + P(e_6) - P(e_4)P(e_6)$$

$$P\left(A \cup B\right) = P(A) + P(B) - P\left(A \cap B\right)$$

$$P\left(A \cup B \cup C\right) = P(A) + P(B) + P(B) + P(A)P(B)P(C) - P(A)P(B) - P(A)P(C) - P(C)P(B)$$



$P(e_i)$, that an event e_i occurs

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“or” equals addition!

$$P(\cup_{i=1}^{\infty} e_i) = \sum_{i=1}^{\infty} P(e_i)$$

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$$P\left(\bigcap_{i=1}^{\infty} e_i\right) = \prod_{i=1}^{\infty} P(e_i)$$

$$P\left(e_4 \cup e_6\right) = P(e_4) + P(e_6) - P(e_4)P(e_6)$$

inclusion - exclusion principle

$$P\left(A \cup B\right) = P(A) + P(B) - P\left(A \cap B\right)$$

$$P\left(A \cup B \cup C\right) = P(A) + P(B) + P(C) - P(A)P(B) - P(A)P(C) - P(B)P(C) + P(A)P(B)P(C)$$

complement probability for not A, $\bar{A}$:

$$P(\bar{A}) = 1 - P(A)$$

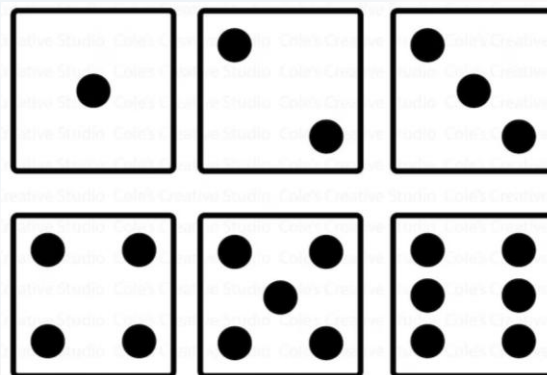
because: $P(e_i) = [0, 1]$ for any e_i



$p(x)$, that an event x occurs

axioms of probability
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the sampling error

discrete (= countable)



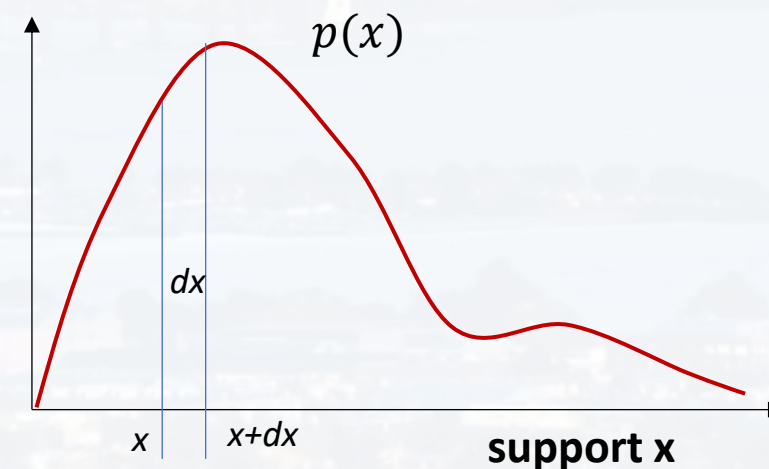
continuous

$[a \leq x \leq b]$

$p(x)$ doesn't make sense

$\rightarrow p(x) dx$

probability **d**ensity **f**unction

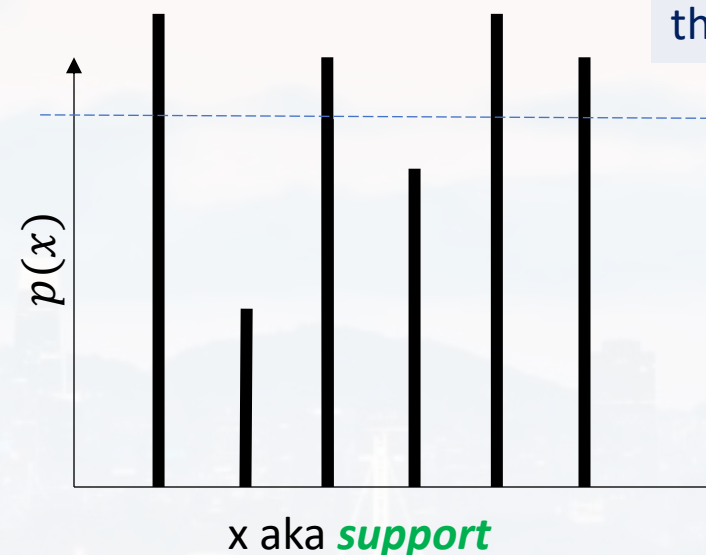
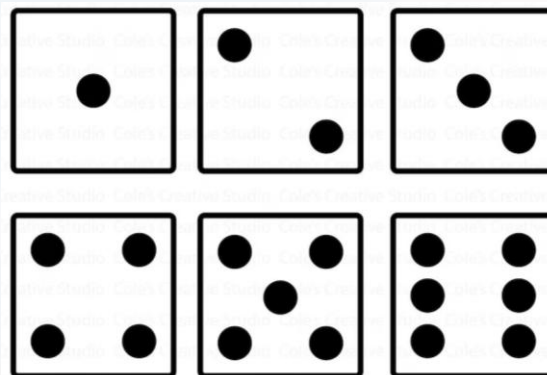




$p(x)$, that an event x occurs

axioms of probability
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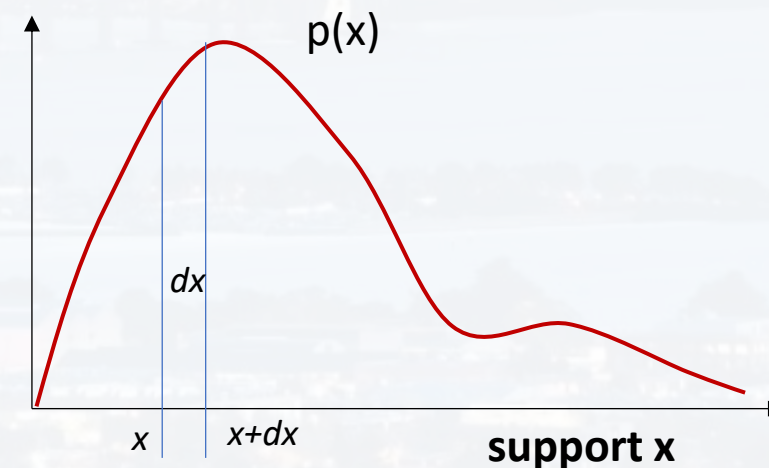
continuous

$[a \leq x \leq b]$

$p(x)$ doesn't make sense

$\rightarrow p(x) dx$

probability **d**ensity **f**unction





$p(x)$, that an event x occurs

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the mean μ

(barycenter)

the variance σ^2

(natural scatter)

discrete (= countable)

$$\mu = E(x) = \sum_i x_i p(x_i)$$

$$\sigma^2 = \text{var}(x) = \sum_i (x_i - \mu)^2 p(x_i)$$

continuous

$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = \text{var}(x) = \int (x - \mu)^2 p(x) dx$$

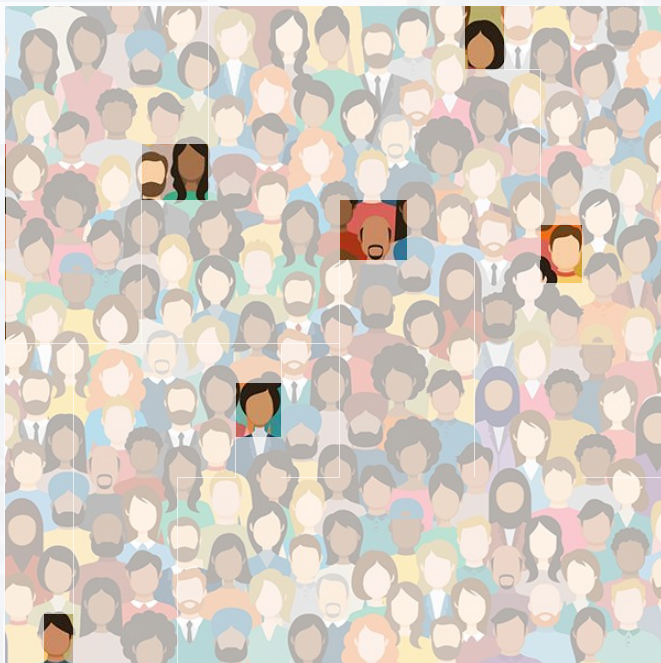


task: you want to find out which **fraction q** of the population doesn't like vanilla ice cream

axioms of probability
important quantities
the sampling error

problem: it is not possible, to ask everyone, so you need to estimate a value for **q , $\bar{q}$** .

solution: you can estimate the error of **$\bar{q}$, $\varepsilon_{\bar{q}}$** as a function of number n of people you asked



all N people

those who don't like vanilla ice cream

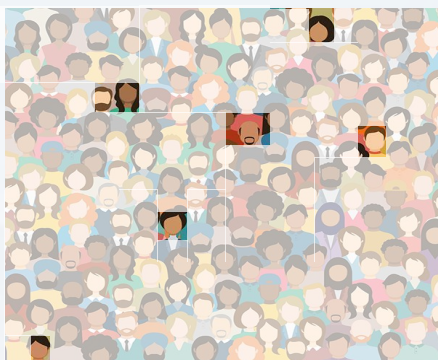
N : population size
 n : sample size



task:

you want to find out which **fraction q** of the population doesn't like vanilla ice cream

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each time you interview a person, (s)he

- doesn't like vanilla ice cream with a probability of **q**
- does like vanilla ice cream with a probability of **$1 - q$**

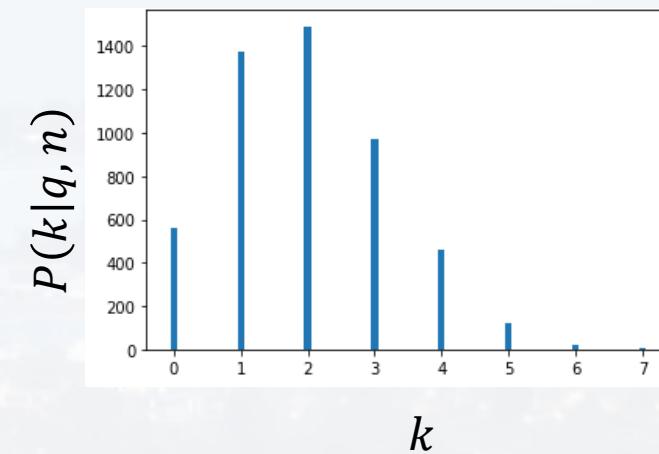
N : population size
 n : sample size

What is the probability $P(k|q, n)$ to find **k** people who don't like vanilla ice cream if you randomly ask **n** people?

answer: it is a **binomial problem**

$$P(k|q, n) = \binom{n}{k} q^k (1 - q)^{n-k}$$

binomial distribution





task: you want to find out which **fraction q** of the population doesn't like vanilla ice cream

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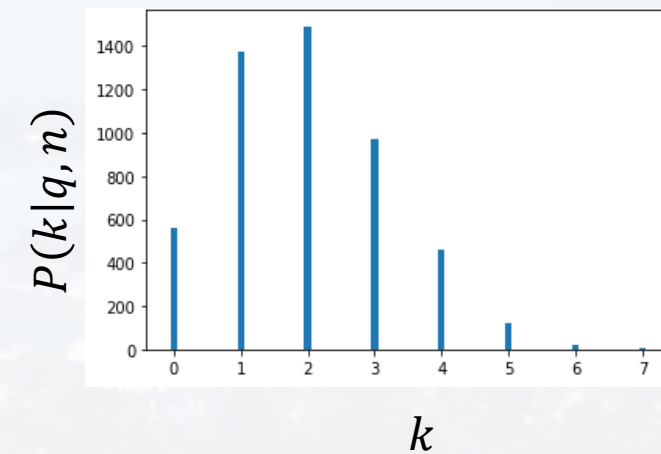
binomial distribution

Each time we'd ask n randomly chosen people, k would be different!

variance of k (binomial)

$$\text{var}(k) = \sum_{k=0}^n (k - qn)^2 \binom{n}{k} q^k (1 - q)^{n-k} = \mathbf{qn(1 - q)}$$

$$\text{var}(x) = \sum_i (x_i - \mu)^2 \mathbf{p(x_i)}$$





task:

you want to find out which **fraction q** of the population doesn't like vanilla ice cream

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$$P(k|q, n) = \binom{n}{k} q^k (1 - q)^{n-k}$$

binomial distribution

N : population size
 n : sample size

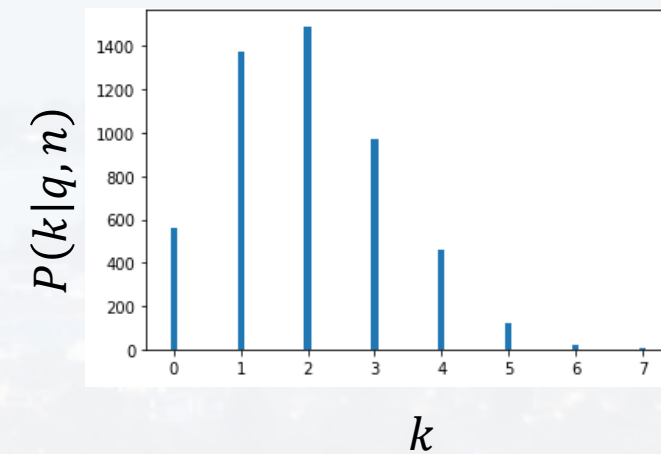
Each time we'd ask n randomly chosen people, k would be different!

$$\text{var}(k) = \mathbf{qn(1 - q)} \quad \sigma_k(k) = \sqrt{qn(1 - q)}$$

our **estimate** for q would be $\bar{q} = \frac{k}{n}$

i. e. the **uncertainty** for q (error propagation!) would be

$$\varepsilon_{\bar{q}} = \frac{\sigma_k(k)}{n} = \frac{\sqrt{qn(1 - q)}}{n} = \frac{\sqrt{q(1 - q)}}{\sqrt{n}} \sim \frac{1}{\sqrt{n}}$$





task:

you want to find out which **fraction q** of the population doesn't like vanilla ice cream

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the **estimate** for q

$$\bar{q} = \frac{k}{n}$$

N : population size
 n : sample size

the **uncertainty** for q would be

$$\varepsilon_{\bar{q}} = \frac{\sqrt{q(1-q)}}{\sqrt{n}} \sim \frac{1}{\sqrt{n}}$$

note:

- equation only applies for $N = \infty$

- one can show for $N \neq \infty$ that $\varepsilon_{\bar{q}} = \frac{\sqrt{q(1-q)}}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$

Finite Population Correction

- **population mean:**

$$\mu = \frac{1}{N} \sum_i^N x_i$$

sample mean:

$$\bar{x} = \frac{1}{n} \sum_i^n x_i$$

- **population variance:**

$$\sigma^2 = \frac{1}{N} \sum_i^N (x_i - \mu)^2$$

sample variance:

$$S^2 = \frac{\sum_i^n (x_i - \bar{x})^2}{n-1}$$

- better way for estimating q or other quantities of the population:

Bayesian Parameter Estimation



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idea: gain of information = “**degree of surprise**”
if something happens that we expected already → no surprise
→ no information

$p(x_i)$: probability of event x_i



$h(x_i)$: function that measures “information”

$$\log[p(x_i) p(x_j)] = \log[p(x_i)] + \log[p(x_j)]$$

- 1) $h(x_i)$ should be additive
- 2) $h(x_i)$ should be monotonic
- 3) if x_i and x_j are independent, then

$$h(x_i, x_j) = h(x_i) + h(x_j)$$

$$p(x_i, x_j) = p(x_i)p(x_j)$$



idea: gain of information = “**degree of surprise**”
if something happens that we expected already → no surprise
→ no information

$p(x_i)$: probability of event x_i

$h(x_i)$: function that measures “information”

- 1) $h(x_i)$ should be additive $h(x_i, x_j) = h(x_i) + h(x_j)$
- 2) $h(x_i)$ should be monotonic
- 3) if x_i and x_j are independent, then $p(x_i, x_j) = p(x_i)p(x_j)$

$$\log[p(x_i) p(x_j)] = \log[p(x_i)] + \log[p(x_j)] \quad p(x_i) \leq 1$$

$h(x_i) = -\log[p(x_i)]$ information is **positive**
low $p(x_i) \rightarrow$ “great surprise”



idea: gain of information = “**degree of surprise**”

if something happens that we expected already → no surprise

→ no information

$$h(x_i) = -\log[p(x_i)]$$

$p(x_i)$: probability of event x_i

The event x_i is randomly drawn from $p(x_i)$

→ **average** amount of information

Entropy S

$$S = - \sum_{i=1}^I p(x_i) \log[p(x_i)] \quad (\text{discrete})$$

$$S = - \int p(x) \log[p(x)] dx \quad (\text{continuous or differential})$$

note: - the **base** of log is **arbitrary** (often 2 or e)

$$- \lim_{p \rightarrow 0} (p \log p) = 0$$

- S is large → no information

- S is zero → all information

- continuous entropy **can** be negative

- continuous entropy is **not** exactly equivalent to discrete entropy (see LDDP)



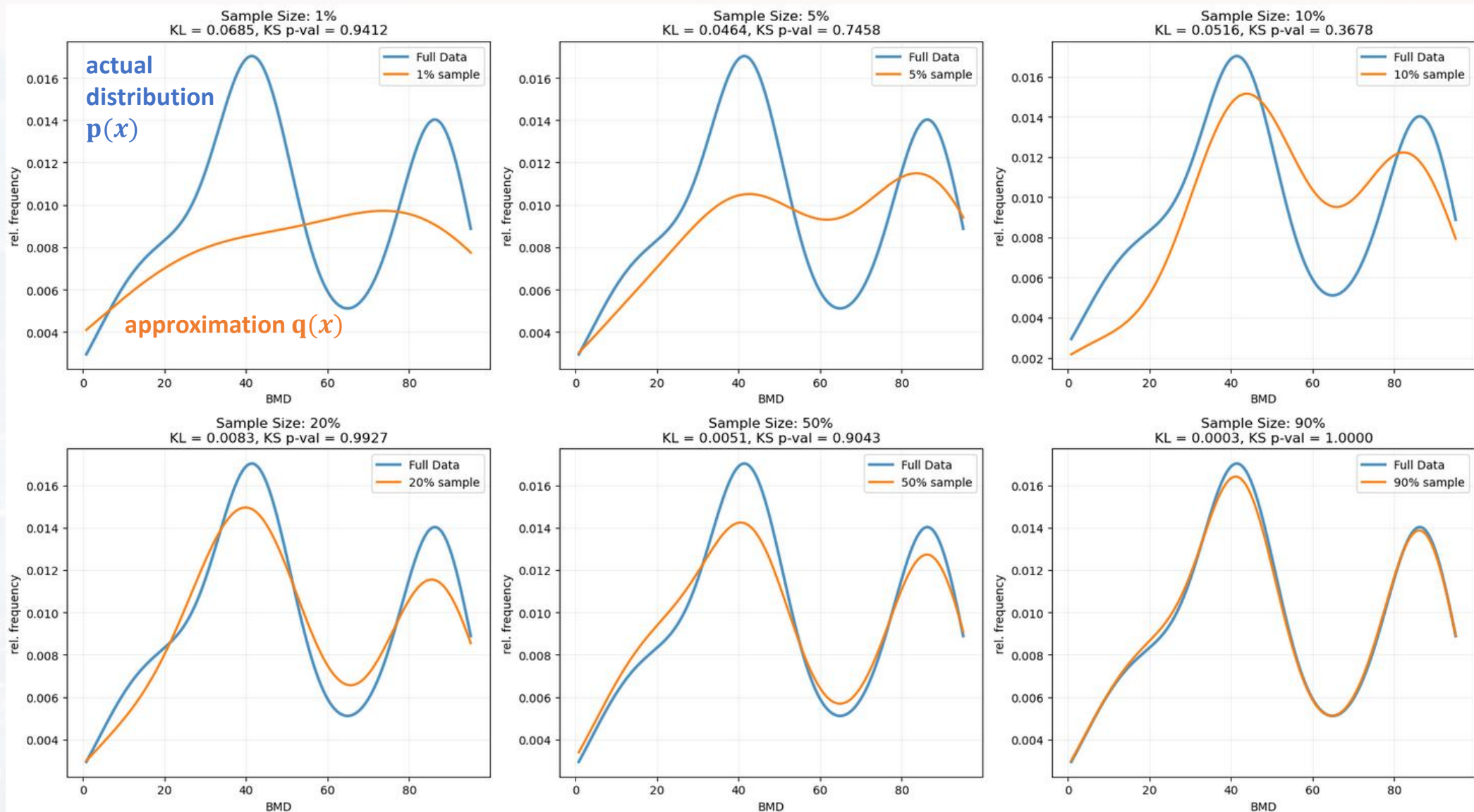
often, we (need to) approximate $p(x)$ by some other distribution $q(x)$



How much is our information “off” if we work with the approximation $q(x)$?



How much is our information “off” if we work with the approximation $q(x)$?





often, we (need to) approximate $p(x)$ by some other distribution $q(x)$



How much is our information “off” if we work with the approximation $q(x)$?

$$-\int p(x) \log[q(x)] dx - \left[-\int p(x) \log[p(x)] dx \right] = -\int p(x) \log \left[\frac{q(x)}{p(x)} \right] dx = KL(p||q)$$

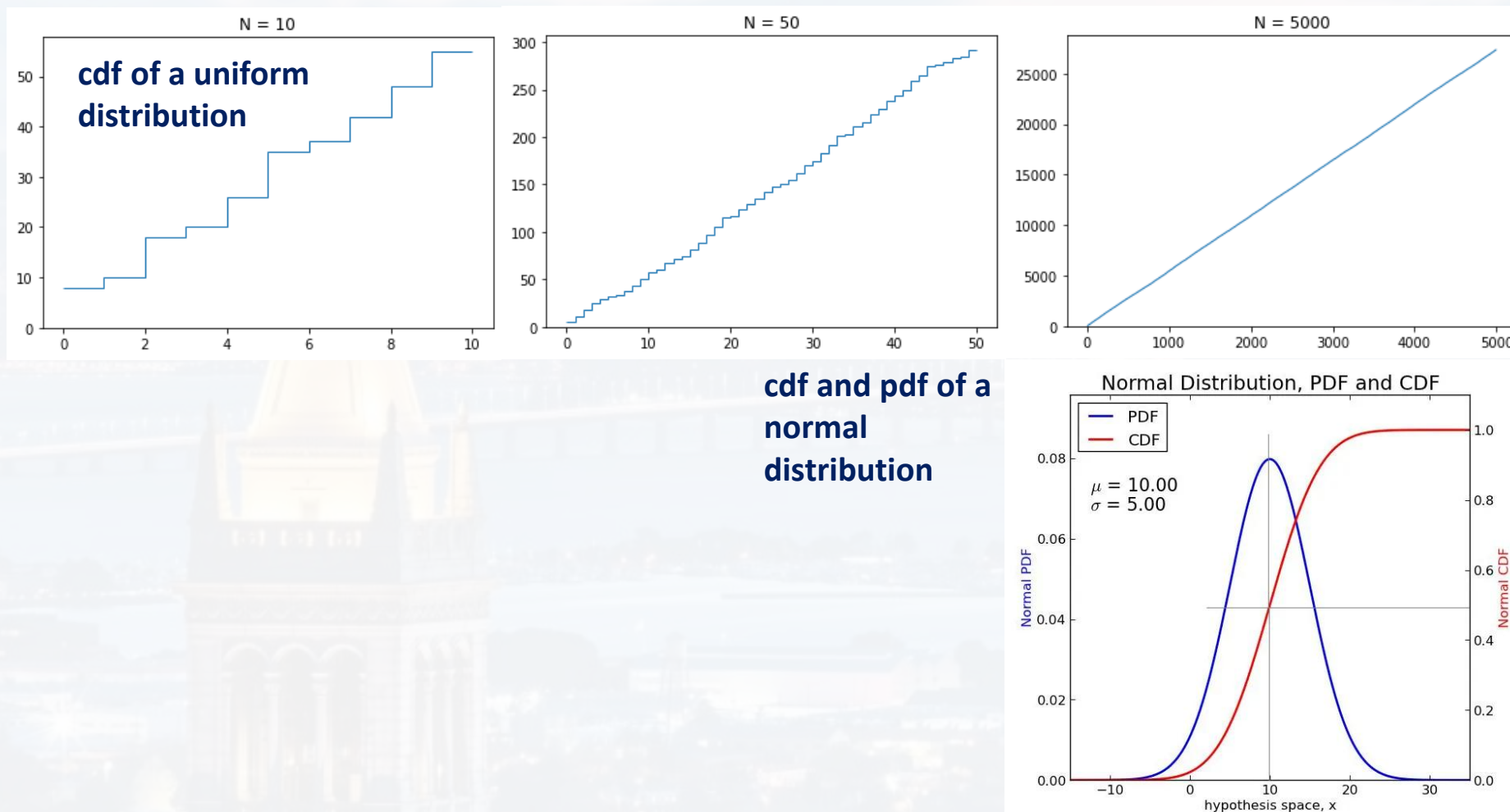
KL or *Kullback-Leiber divergence*

- is **not** a distance! $KL(p||q) \neq KL(q||p)$
- ranges from zero (both distributions are equal) to ∞



another way to compare distributions is the **Kolmogorov – Smirnov – test (KS test)**

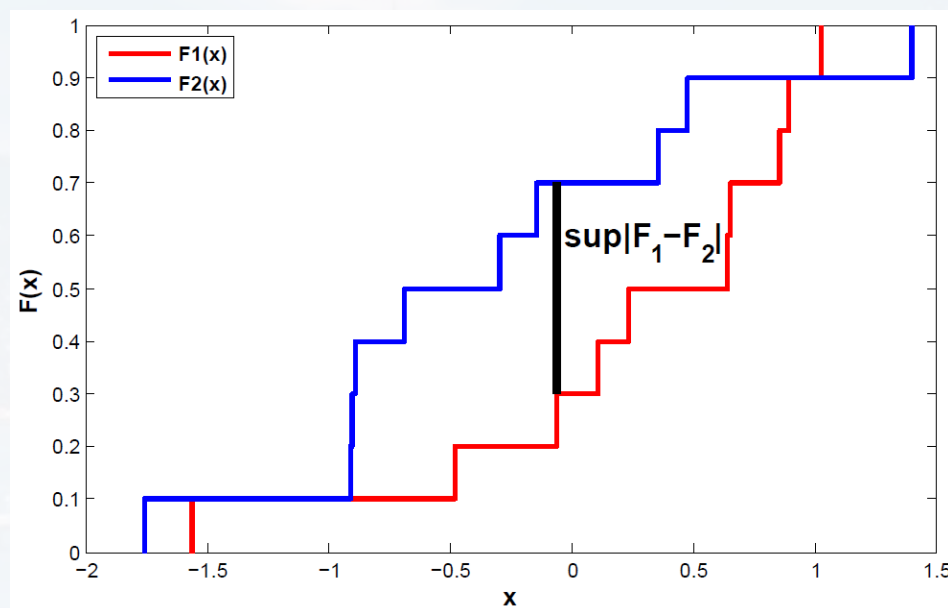
comparing the **Cumulative density functions** $P(x) = \int_a^x p(x) dx$





another way to compare distributions is the **Kolmogorov – Smirnov – test (KS test)**

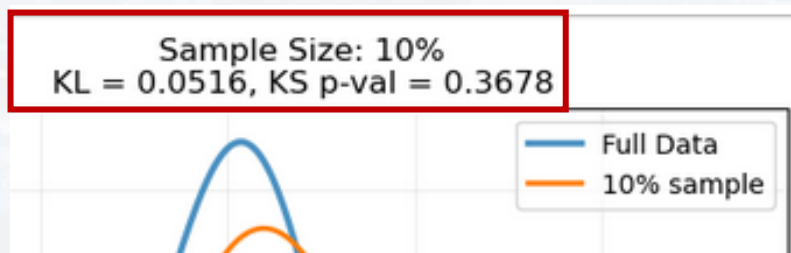
comparing the **Cumulative density functions** $P(x) = \int_a^x p(x) dx$



KS test compares the cdfs F_1 and F_2 of two pdfs by looking at the largest difference D

→ returns a p-value:

*“The probability that we obtain such a **value (or larger) for D** , assuming **F_1 and F_2** are obtained from the **identical pdf!**”*





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data
set

group I	data points			

group II			
----------	--	--	--

group III	data points			

- groups can differ in sizes
- data is usually sign. different between groups

Random Sampling

Stratified Sampling
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	TARGET_LEVEL	HIT_CALL	BMD
group I	gene	0.988987	46.472969
	gene	0.962960	42.798135
	gene	0.917757	41.108881
group II	cell	0.960401	78.170916
	cell	0.957572	89.467940



how:

- sampling regardless of groups and their proportions

pros:

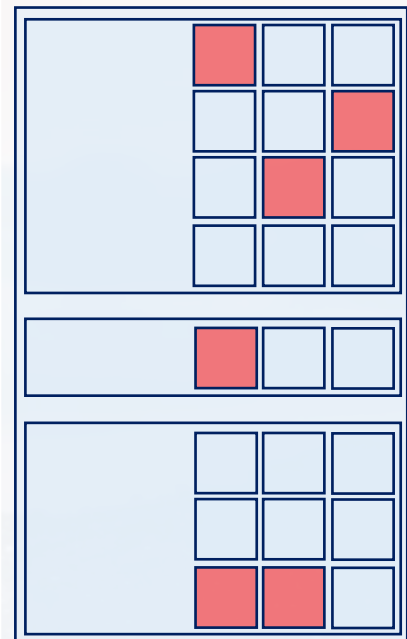
- simple
- unbiased
- automatically maintains *approx.* proportion of groups

cons:

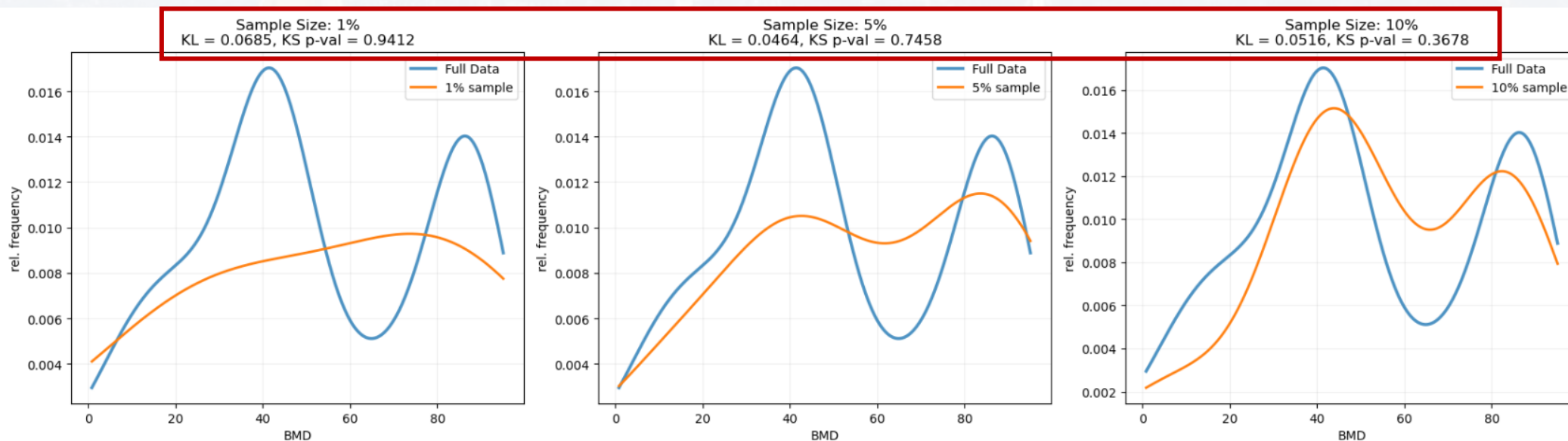
- needs large sample size, especially if many groups with different proportions
- list of groups need to be complete

Random Sampling

Stratified Sampling
Systematic Sampling
Cluster Sampling
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Oversampling & Undersampling



N = 420



see: `SamplingMethods.ipynb`



how:

- sampling regardless of groups and their proportions

pros:

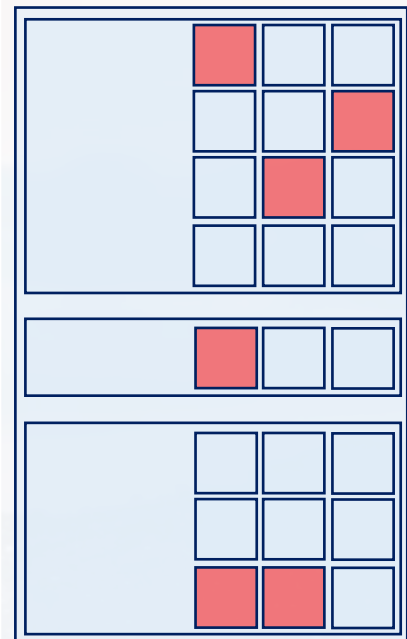
- simple
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- automatically maintains *approx.* proportion of groups

cons:

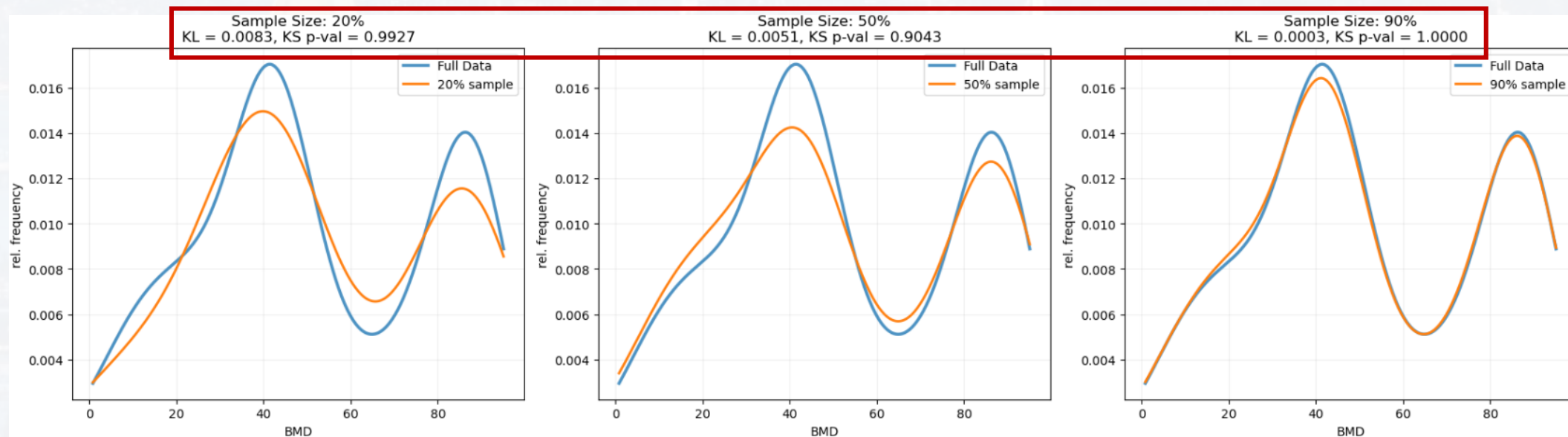
- needs large sample size, especially if many groups with different proportions
- list of groups need to be complete

Random Sampling

Stratified Sampling
Systematic Sampling
Cluster Sampling
Bootstrap Sampling
Oversampling & Undersampling



N = 420



see: SamplingMethods.ipynb



Random Sampling
Stratified Sampling
Systematic Sampling
Cluster Sampling
Bootstrap Sampling
Oversampling & Undersampling

how:

- sampling within groups (aka **strata**)
- population is exhaustively partitioned into **disjoint subgroups**

pros:

- maintains *exact*. proportion of groups
- useful when groups are expected to follow different distributions

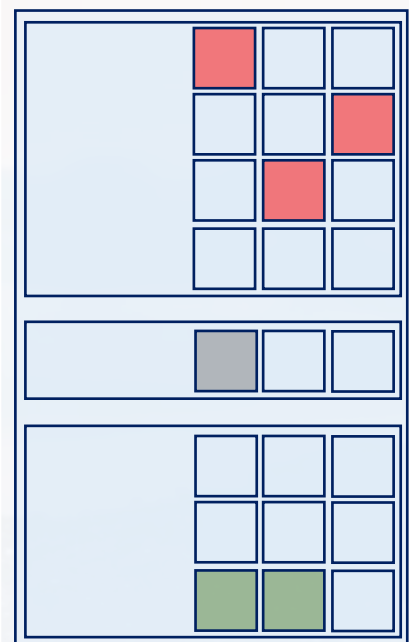
cons:

- don't apply if the population cannot be exhaustively partitioned into disjoint subgroup

TARGET_LEVEL	HIT_CALL	BMD
gene	0.988987	46.472969
gene	0.962960	42.798135
gene	0.917757	41.108881
cell	0.960401	78.170916
cell	0.957572	89.467940

```
L = list(File[col_groups])
for g in Groups:
    ct = L.count(g)
    print(g + ": " + str(ct) + " appearances" )
```

```
organ: 102 appearances
none: 3 appearances
tissue: 18 appearances
gene: 202 appearances
cell: 61 appearances
chemical: 14 appearances
random: 20 appearances
```



how:

- sampling within groups (aka **strata**)
- population is exhaustively partitioned into **disjoint subgroups**

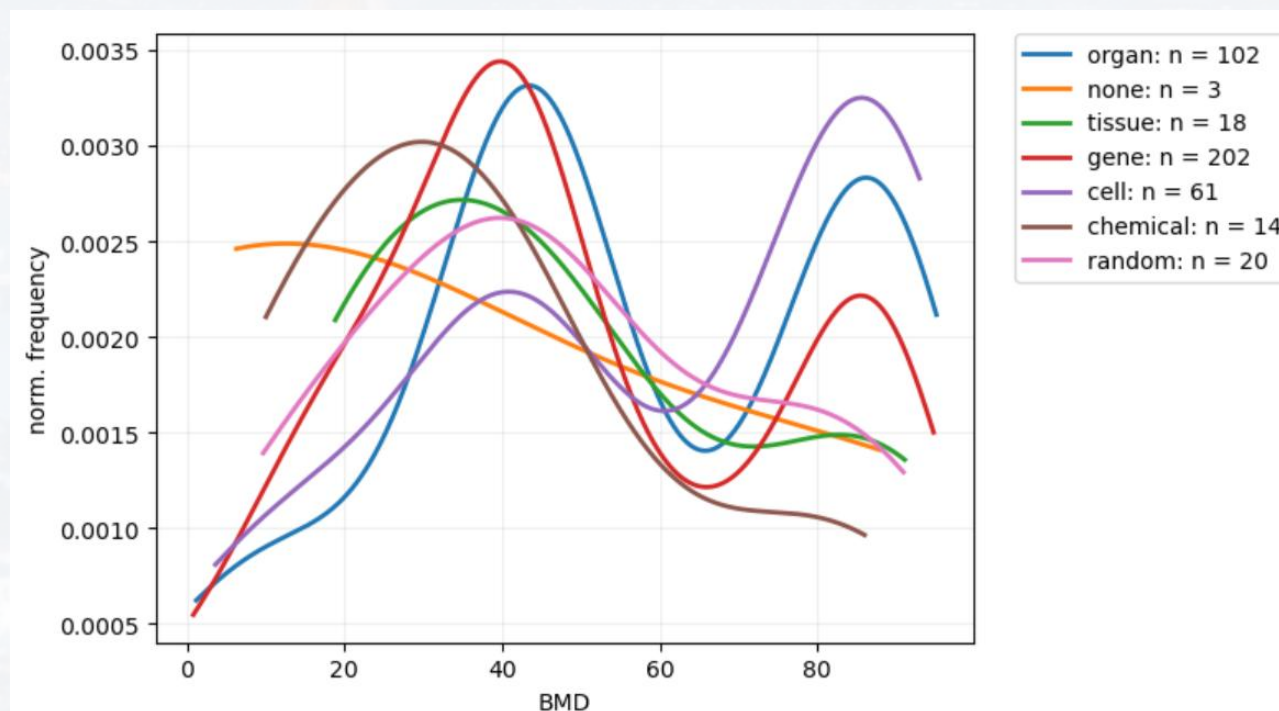
pros:

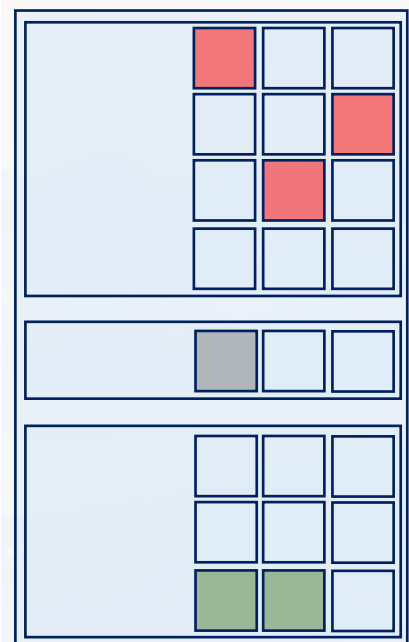
- maintains *exact*. proportion of groups
- useful when groups are expected to follow different distributions

cons:

- don't apply if the population cannot be exhaustively partitioned into disjoint subgroup

Random Sampling
Stratified Sampling
Systematic Sampling
Cluster Sampling
Bootstrap Sampling
Oversampling & Undersampling





how:

- sampling within groups (aka **strata**)
- population is exhaustively partitioned into **disjoint subgroups**

pros:

- maintains *exact*. proportion of groups
- useful when groups are expected to follow different distributions

cons:

- don't apply if the population cannot be exhaustively partitioned into disjoint subgroup

Random Sampling
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drawing 20% from each group

organ: 102 appearances in full data set,

none: 3 appearances in full data set,

tissue: 18 appearances in full data set,

gene: 202 appearances in full data set,

cell: 61 appearances in full data set,

chemical: 14 appearances in full data set,

random: 20 appearances in full data set,

20 appearances in subsample.

1 appearances in subsample.

4 appearances in subsample.

40 appearances in subsample.

12 appearances in subsample.

3 appearances in subsample.

4 appearances in subsample.

Ratio = 0.20

Ratio = 0.33

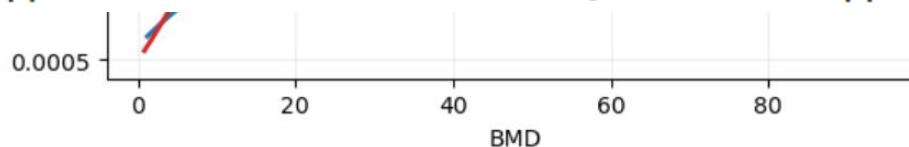
Ratio = 0.22

Ratio = 0.20

Ratio = 0.20

Ratio = 0.21

Ratio = 0.20





how:

- sampling each k-th element

pros:

- simple
- samples evenly over population

cons:

- imposes bias if there are hidden pattern in the data

Random Sampling
Stratified Sampling
Systematic Sampling
Cluster Sampling
Bootstrap Sampling
Oversampling & Undersampling

0	46.472969
20	88.885569
40	63.151552
60	26.627395
80	35.520452
100	19.830881
120	38.801088
140	43.063503
160	89.690644
180	85.789285
200	89.592299
220	86.492331
240	41.470968
260	13.242978
280	85.977758
300	62.225787
320	85.612638
340	16.862983
360	85.480457
380	27.196121
400	22.218282



Random Sampling
Stratified Sampling
Systematic Sampling
Cluster Sampling
Bootstrap Sampling
Oversampling & Undersampling

how:

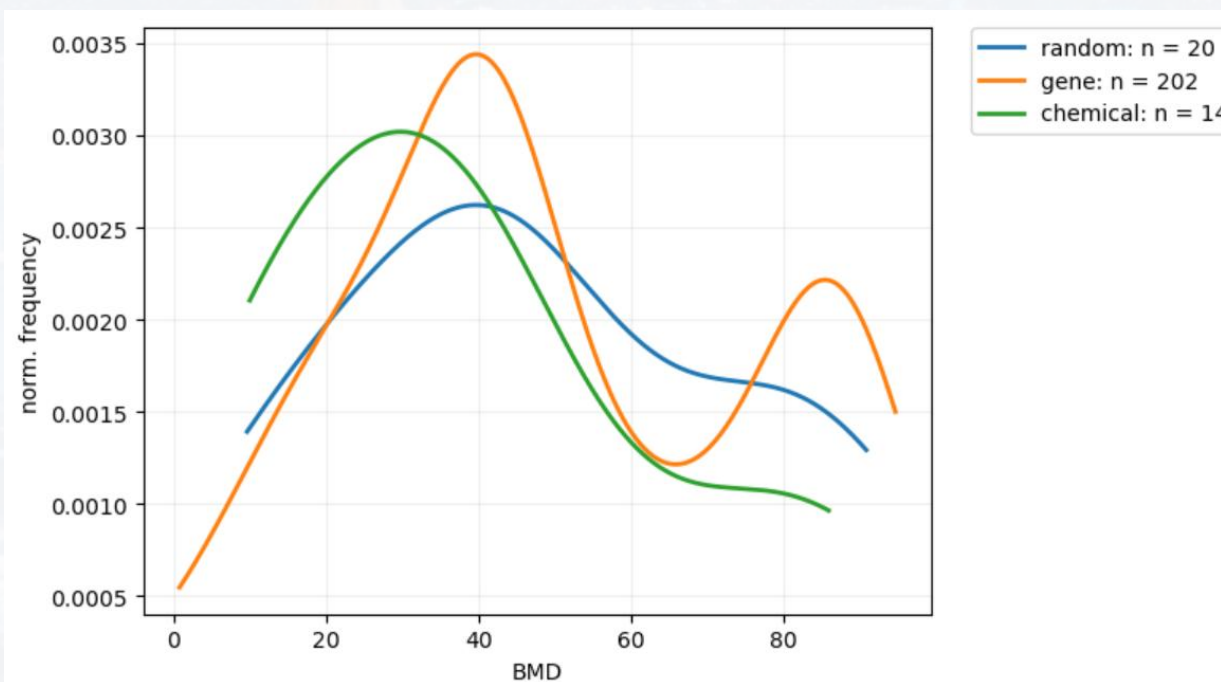
- randomly select the groups (aka cluster)
- sample from those cluster

pros:

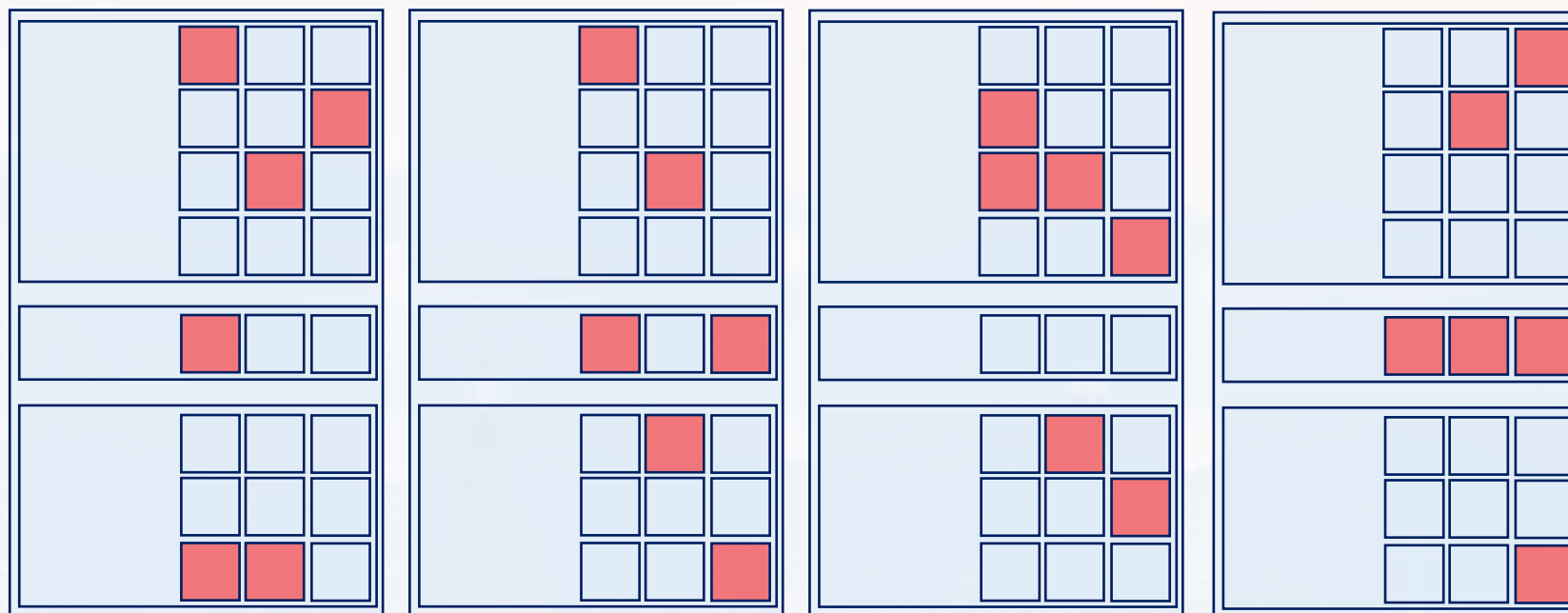
- efficient **for large data sets** (eg. polls)
- if cluster not expected to differ a lot

cons:

- prone to bias
- prone to sample errors
- only use if no other choice!



- randomly select three groups
- draw 20% from each group



- Random Sampling
- Stratified Sampling
- Systematic Sampling
- Cluster Sampling
- Bootstrap Sampling**
- Oversampling & Undersampling

how:

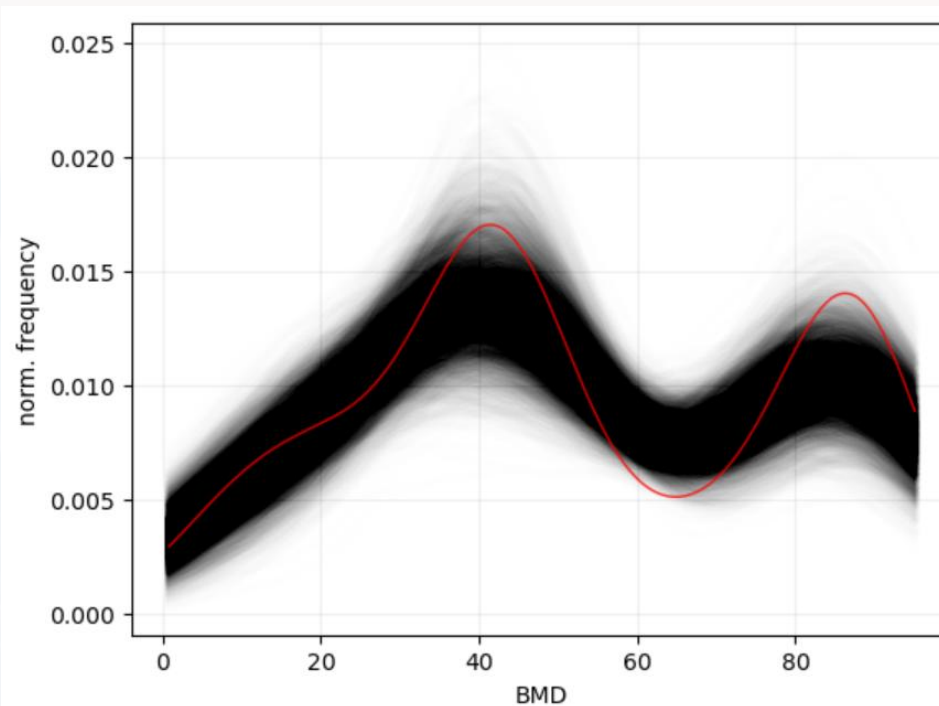
- sampling with replacement, N_{boot} times

pros:

- simple
- for estimating sample error empirically → percentiles

cons:

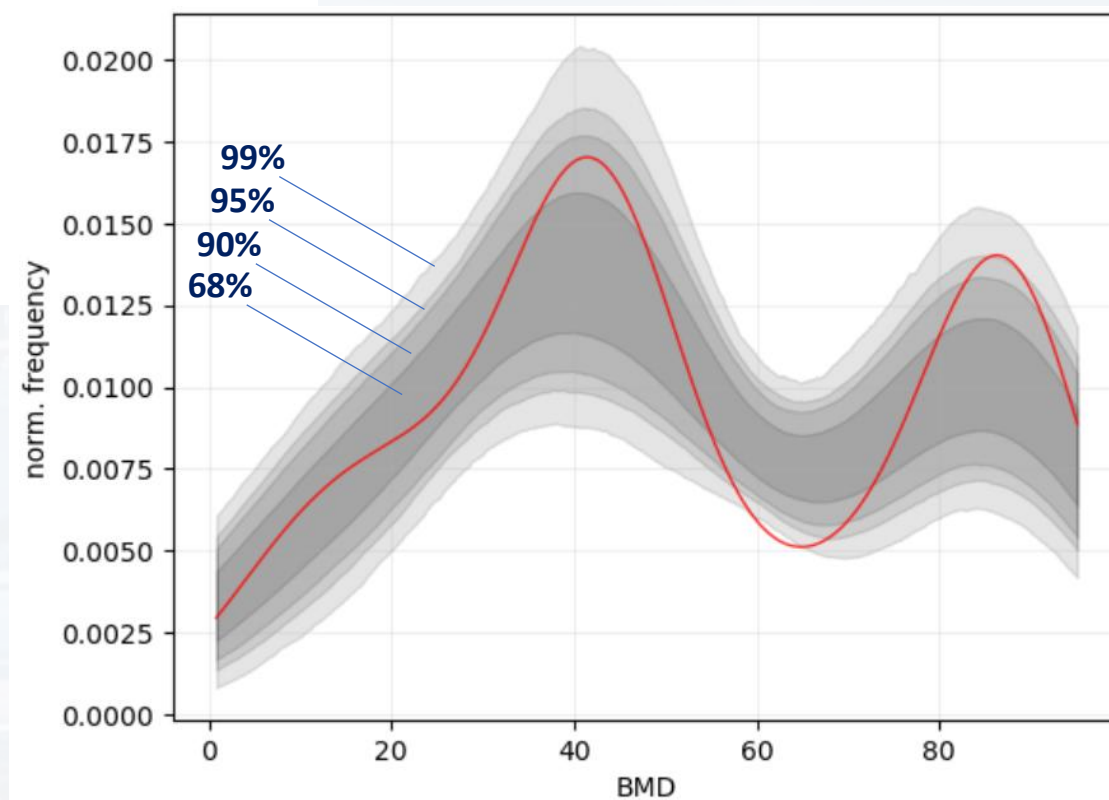
- $N_{boot} \gg n$



— Full Data (size 420)

- $n = 50$
- $N_{boot} = 5000$

Random Sampling
Stratified Sampling
Systematic Sampling
Cluster Sampling
Bootstrap Sampling
Oversampling & Undersampling



see: `SamplingMethods.ipynb`



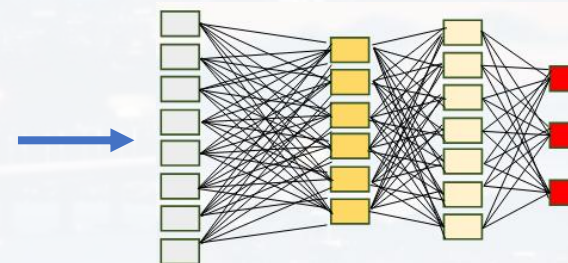
- how:** - sampling according to group imbalance
- pros:** - compensates sample imbalance
- cons:** - repetitions

Random Sampling
Stratified Sampling
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Cluster Sampling
Bootstrap Sampling
Oversampling & Undersampling

9,900 dog images



100 cat
images



dog!!

99% accuracy



how: - sampling according to group imbalance

pros: - compensates sample imbalance

cons: - repetitions

Random Sampling
Stratified Sampling
Systematic Sampling
Cluster Sampling
Bootstrap Sampling
Oversampling & Undersampling

9,900 dog images



100 cat
images



	x	y
178	-1.680904	cat
113	-0.442641	cat
37	0.606229	cat
178	-1.680904	cat
160	-1.026721	cat
...	...	...
66	0.415702	cat
66	0.415702	cat
83	2.124732	cat
101	-0.686784	cat
83	2.124732	cat



how: - sampling according to group imbalance

pros: - compensates sample imbalance

cons: - repetitions

Random Sampling
Stratified Sampling
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Oversampling & Undersampling

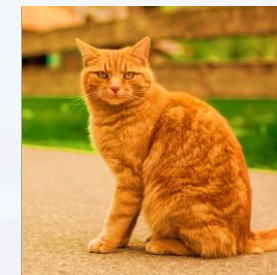
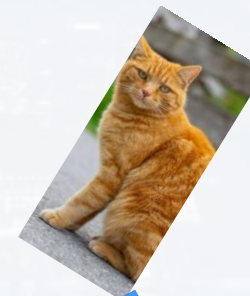
9,900 dog images



100 cat
images



data augmentation for reducing repetitions!





Thank you very much for your attention!

